Unforgeable Process Evidence: A New Paradigm of Scientific Integrity Complementing Reproducibility

Chong Liu*

Manuscript version 0.2.0 — August 14, 2026

Abstract

Reproducibility has become the dominant frame for scientific integrity: a result is trusted insofar as an independent party can rebuild it from shared data and code. We argue that reproducibility verifies *what* was claimed but remains silent about *how* the claim was arrived at, and that this second question is a distinct and complementary axis of integrity that the present crisis exposes without resolving. We propose *unforgeability* as the operative property of that axis: evidence about the research process that is computationally infeasible to fabricate or retroactively alter, so that a validated record is itself non-trivial evidence that the reported process occurred as and when it is attested to have occurred. We develop the scientific rationale for treating unforgeable process evidence as a new paradigm of scientific integrity—a support for truth-claims that complements, rather than replaces, reproducibility—and we show that the property is technically achievable today with local, privacy-preserving, cryptographic recording that any third party can verify offline. We instantiate the idea in a concrete exemplar: the open Evidence Package v1 format and its Rust/Tauri reference implementation (Academic Integrity Recorder), which maintains a hash-chained, device-signed, encrypted record and discloses its own capability boundaries honestly. Throughout, we are careful about what unforgeability does *not* establish: it attests process integrity, not truth, originality, authorship, or completeness. We close with scope, limitations, and a governance path for adoption.

1 Introduction

The reproducibility crisis is usually told as a story about *results*: too many published findings cannot be rebuilt from the materials that authors share [Nosek et al., 2015, Stodden, 2009]. The standard remedy has therefore been to share more—data, code, protocols, preregistrations—so that an independent party can re-derive the outcome [Nosek et al., 2015, Stodden, 2009]. This framing has made *reproducibility* the de facto currency of scientific integrity: a result is trustworthy to the extent that it can be regenerated.

We propose a complementary frame. Reproducibility answers the question *"can the result be rebuilt?"* It does not answer *"was the result arrived at honestly?"* A finding can be perfectly reproducible and yet be the product of selective reporting, post-hoc hypothesizing, undisclosed reuse of a collaborator's work, or an AI-assisted draft presented as original. Conversely, a genuinely honest workflow may fail to reproduce for reasons unrelated to misconduct—fragile materials, unreported

*Corresponding author. Independent Researcher. ORCID: 0009-0009-8116-1255. Email: chong.liu.phil@outlook.com. Code and specification: `https://github.com/chongliuresearch/academic-integrity-recorder`.

environmental conditions, or legitimate stochasticity [Ioannidis, 2005]. The two questions are correlated but distinct, and the crisis literature documents the blind spot explicitly: even when a study is replicated, an observer cannot tell whether the *process* that produced the original was faithful [John et al., 2012, Fanelli, 2009].

What is missing, we argue, is not another call for transparency but a precise, checkable property. We call that property *unforgeability*. A process-evidence record is unforgeable if it is computationally infeasible—under explicitly stated assumptions—to produce a record that validates for a process that did not actually occur, or to alter, reorder, or backdate an existing record without detection. Under this property, successful validation of a record becomes non-trivial evidence that the recorded process happened as attested. Unforgeability is thus not an assertion of virtue but a conditional, mechanistic guarantee: it narrows the set of ways a claim could be false by removing those that depend on fabricating the process itself. It serves the broader goal of *process trustworthiness*—verifiable evidence that the conduct of research was recorded as it happened—by giving that goal a formal, falsifiable core.

The distinction between the goal and the property matters. "Process trustworthiness" describes an aspiration that is easy to affirm and hard to evaluate; "unforgeability" names a property that a skeptic can actually check, because checking it reduces to re-deriving hashes and signatures. A design that aims at unforgeability commits itself to concrete mechanisms—an append-only hash chain, canonical serialization, a device signature, and an external timestamp—each of which can fail or be attacked, and each of which can therefore be stated and scrutinized. This is the sense in which the concept is scientifically productive rather than merely rhetorical.

Our contributions are:

1. a clear separation of *reproducibility* (what was claimed) from *unforgeable process evidence* (how the claim was arrived at), and an argument that the latter constitutes a new paradigm of integrity complementing reproducibility;

2. a characterization of unforgeability for research-process evidence, stated as a conditional guarantee under explicit cryptographic and operational assumptions, together with its epistemic role as a complement to reproducibility;

3. an Evidence Package v1 specification and an open reference implementation (Academic Integrity Recorder) in Rust (core), Tauri (desktop), and browser/VS Code/shell extensions, with an offline verifier, demonstrating that the property is achievable with local, privacy-preserving tooling;

4. a candid treatment of scope, adversarial limits, and a governance path for adoption.

Non-claim. Unforgeable process evidence proves process integrity—that the recorded process occurred as recorded, on the attested device, at the attested time, modulo the stated cryptographic assumptions. It does *not* certify identity, authorship, originality, correctness, or academic integrity, and it cannot prove that all research activity was captured. This boundary is enforced in code and in the exported manifest, not merely asserted in prose.

2 The Reproducibility Crisis

2.1 What the empirical record shows

The crisis is not anecdotal. In a 2016 *Nature* survey of 1,576 researchers, **70%** reported having failed to reproduce another group's experiment, and more than half reported failing to reproduce *their* *own* [Baker, 2016]. Among large coordinated replications, the Open Science Collaboration reran 100 psychology studies and found only **36%** reproduced the original statistically significant result, with effect sizes roughly half as large [Open Science Collaboration, 2015]. In preclinical cancer biology, the Reproducibility Project reproduced only a minority of high-impact findings [Errington et al., 2021], consistent with earlier industry reports that roughly **11%** (6 of 53) of landmark preclinical studies could be reproduced internally [Begley and Ellis, 2012] and that a major pharmaceutical pipeline observed reproducibility in only about one fifth of projects [Prinz et al., 2011].

These numbers are not a verdict on any discipline's honesty; they are a verdict on the *visibility* of process. As Ioannidis [2005] argued two decades ago, the prevailing incentive structure predicts a large fraction of false positives even without malfeasance, and John et al. [2012] documented that questionable research practices (p-hacking, selective reporting, undisclosed flexibility) are common and often unconscious.

2.2 Why reproducibility alone is insufficient

Reproducibility operates on the *output* side of research. It asks whether, given shared inputs, the same outputs reappear. Three gaps follow.

First, **the reporting gap**. Reproducibility depends on what authors choose to share. Studies show extensive under-disclosure of experimental design, and that the absence of a disclosure norm is itself a cause of non-reproduction [Nosek et al., 2015]. What is never written down cannot be reproduced, and the decision of what to write down is exactly the process reproducibility cannot see.

Second, **the selectivity gap**. Even with full data, reproducibility cannot distinguish a result obtained by pre-specified analysis from one obtained by exploring many analyses and reporting the best [Ioannidis, 2005, Gelman and Loken, 2014]. Preregistration addresses this prospectively [Nosek et al., 2018], but it covers a narrow slice of ordinary research behavior and is silent about the long, messy trail of reading, drafting, and revision that precedes any registration.

Third, **the provenance gap**. Modern research increasingly involves tools—search, writing environments, version control, and AI assistants—whose use is rarely disclosed in a verifiable way [Gao et al., 2023]. Whether and how an AI contributed to a manuscript, whether a passage was copied from a collaborator, or whether a figure was regenerated after peer review are questions about *process*, and reproducibility is structurally blind to them.

None of this diminishes reproducibility; it motivates a second pillar that observes the conduct of research directly rather than inferring it from its outputs.

3 Unforgeability: Rationale and Epistemic Role

3.1 The warrant of a scientific claim

A scientific result functions as evidence for a claim about the world to the degree that the process producing it can be trusted. That trust decomposes into two questions that are easy to conflate but distinct in what they can be checked. The first asks whether the reported result follows from the reported materials—call this the question of *reproducibility*, of *what* was claimed. The second asks whether the reported process is the *actual* process—whether the experiment, the reading, the analysis, and the writing happened as they are described, in the order described, and before the outcome was known. Call this the question of *fidelity*, of *how* the claim was arrived at. A result can score high on the first question and low on the second, and vice versa.

Science has always treated the second question as load-bearing, even when it lacked the machinery to answer it mechanically. The laboratory notebook is a centuries-old technology whose entire purpose is to make the conduct of experiment visible to later scrutiny, precisely because its contemporaneity—the fact that its entries were written as events unfolded rather than reconstructed afterward—is what gives it evidentiary weight [Macrina, 2017]. The W3C PROV model formalizes provenance as a first-class, machine-checkable object [World Wide Web Consortium, 2013, Moreau and Missier, 2013, Moreau and Groth, 2015], and open-science movements have long insisted that transparency of process is as important as availability of products [Nosek et al., 2015, Fanelli, 2011]. What these strands share is an implicit commitment to exactly the property we wish to make explicit: that the account of how a result was produced should itself be trustworthy.

The reproducibility crisis can be read as the failure of the present ecosystem to secure this second question. Its signature behaviors—p-hacking, HARKing, selective reporting, the post-hoc construction of a clean narrative—do not primarily attack the regenerability of a result; they attack the fidelity of its provenance, and they survive reproduction precisely because a fabricated-but-consistent artifact can be re-derived just as easily as a genuine one. Reproducibility, by construction, cannot tell the two apart: it verifies the artifact, not the act. If the act is what matters for evidentiary weight, then a property that binds the act—rather than the artifact—is what is missing.

3.2 Unforgeability as the operative property

We therefore need a property that speaks to the act rather than the artifact. We propose *unforgeability*. Informally, a process-evidence record is unforgeable if a would-be fabricator cannot produce a record that passes validation for a process that did not occur, nor alter an existing record—by inserting, deleting, reordering, or backdating events—without that change being detected. The force of the definition lies in what validation then licenses: if a record validates, then, under the assumptions on which the property rests, the recorded process did in fact occur as recorded.

Two qualifications keep this claim honest. First, unforgeability here is *computational* and *conditional*, not absolute. It rests on the hardness of standard cryptographic primitives—collision-resistant hashing, unforgeable signatures under the Ed25519 scheme [Bernstein et al., 2012], and authenticated encryption [Bernstein, 2008]—and on operational assumptions such as the secrecy of signing keys and the integrity of any external timestamp. An adversary who compromises a key, or who controls the clock, can defeat the property, and no honest account can pretend otherwise. Second, unforgeability is a property of the *record*, not a verdict on the *research*. An unforgeable record of a flawed experiment is still a flawed experiment. What the property does is eliminate an entire class of failures—those that operate by fabricating or rewriting the process itself—leaving the remaining uncertainty to methodology and interpretation, which is precisely where peer review and replication are effective.

It is useful to contrast unforgeability with the weaker, more familiar notion of *tamper-evidence*. A tamper-evident record is one whose alteration is detectable; that is a property of the mechanism one uses to build the record. Unforgeability is the property of the record one wants: that no convincing counterfeit can be produced at all. Tamper-evidence is the means; unforgeability is the end. Leading with the end clarifies both the design goal and the security claim, and it is why we organize the exemplar in §4 around achieving unforgeability through a hash chain, canonical serialization, a device signature, and a timestamp—rather than merely around detectable tampering.

3.3 Complementarity with reproducibility

Unforgeability is not a substitute for reproducibility; the two secure different questions, and both are needed. The relationship is cleanest as a two-by-two space (Table 1). A result that is reproducible but whose process is forgeable can be rebuilt yet may rest on a fabricated or selectively edited account. A result whose process is unforgeably recorded but which fails to reproduce points to a problem of method or robustness rather than of honesty—its process is attested, its inference is doubtful. A result that is both reproducible and unforgeably recorded enjoys the strongest available evidentiary basis short of independent corroboration of the underlying phenomenon. And a result that is neither commands the least.

	Reproducible	**Not reproducible**
Unforgeable process	result regenerates *and* process is attested (strongest)	process attested but result does not regenerate (method or robustness doubt)
Forgeable process	result regenerates but process may be fabricated or edited	neither regenerable nor attested (weakest)

Table 1: Unforgeability and reproducibility secure different questions and are independent. Each cell indicates what the two properties jointly establish.

Two observations follow. First, the properties are *independent*: knowing that a result reproduces tells us nothing about whether its process was genuine, and knowing that its process was unforgeably recorded tells us nothing about whether the result will reproduce. This independence is exactly why unforgeability contributes evidence that reproducibility cannot supply. Second, the value of unforgeability is asymmetric and incremental: it does not raise the ceiling of what science can claim, but it raises the floor below which evidence cannot easily fall. By making the fabrication of process computationally infeasible and its detection mechanical, it converts a category of integrity failures from matters of contested testimony into matters of checkable fact. That is a modest but genuine strengthening of the evidentiary basis on which scientific truth-claims rest—modest, because it never establishes truth; genuine, because it narrows the ways a claim could be false.

3.4 Technical feasibility

The property would be idle if it were not attainable with ordinary tooling, on ordinary machines, without imposing surveillance. Three developments make unforgeable process evidence practical now where it was not a decade ago.

Local-first capture without a panopticon. The evidence can be collected on the researcher's own machine, with no server, no telemetry, and sensitive content never leaving the device in cleartext. This addresses the principal objection to process recording—that it is surveillance—and aligns with data-minimization norms [Voigt and Von dem Bussche, 2017]. Local-first is a design choice in service of unforgeability's adoption, not a headline property: it removes the single largest reason a researcher would refuse the tool, and thereby widens the conditions under which an unforgeable record can exist at all.

Cheap, standard cryptography. Every primitive the property needs is library-grade. An append-only hash chain commits each event to its predecessor, so deletion, reordering, or editing breaks the chain [Crosby and Wallach, 2009]. Canonical serialization (RFC 8785 JCS) makes hashing independent of key order and whitespace [Rundgren, 2020]. A device signature under Ed25519 attests the recorder and gives the record its unforgeable anchor [Bernstein et al., 2012]. Authenticated encryption (XChaCha20-Poly1305) seals sensitive content [Bernstein, 2008]. An internet-independent timestamp, such as OpenTimestamps, can later prove existence at or before a point in time without trusting a notary [OpenTimestamps, 2018]. None of this is exotic; all of it is battle-tested.

Offline, adversarial verification. Because the property is mathematical, a skeptic needs no special access to check it. The verifier is a small, auditable program that re-derives every hash and signature; its success, rather than any party's assertion, is what licenses the inference that the record is genuine. This is the same principle that makes signed commits and checksums trustworthy without a trusted third party, and it is what distinguishes unforgeable process evidence from a self-reported methods section, which remains forgeable however detailed it is.

4 A Proposed Exemplar: Evidence Package v1 and Reference Implementation

We present ACADEMIC INTEGRITY RECORDER, an open-source implementation of the above, and the EVIDENCE PACKAGE V1 it emits. The design obeys five principles: *unforgeability*, *local-first capture*, *honest capability disclosure*, *privacy-by-default*, and *offline verification*.

4.1 Architecture

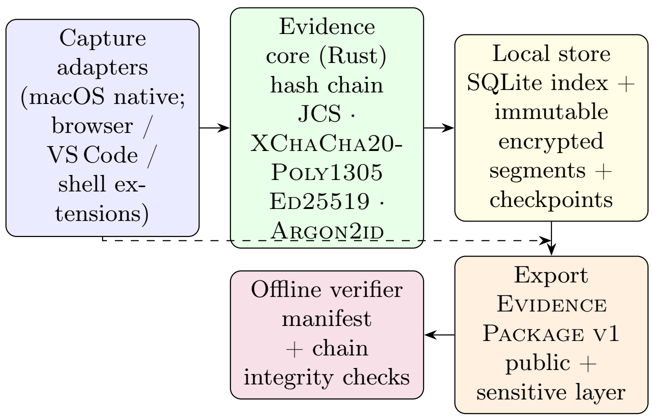

Figure 1: Architecture for unforgeable process evidence. Evidence flows from platform capture adapters into a Rust evidence core that maintains a hash-chained, device-signed record, is persisted in an immutable local store, exported as a two-layer EVIDENCE PACKAGE V1, and verified entirely offline. No telemetry leaves the device.

Capture adapters observe activity—foreground application, file events, browser and editor interactions, shell commands (opt-in), and screenshots of the selected window on supported platforms. They forward events to the evidence core, which canonicalizes, hashes, signs, and persists them (Figure 1).

4.2 The evidence event and hash chain

Each recorded moment is an *evidence event* with a monotonic sequence number, occurrence and capture timestamps, a source, a semantic kind (e.g., input, file-modified, web-interaction, annotation, anchor-created, AI-disclosure), a sensitivity label, and a payload. The core computes

$$\begin{aligned} h_{\text{payload}} &= \text{SHA256}(\text{RFC 8785 JCS}(\text{payload})), \\ h_{\text{event}} &= \text{SHA256}(\text{RFC 8785 JCS}(\text{material})), \end{aligned}$$

where material binds the event's identity, project, timestamps, source, kind, sensitivity, h_{payload}, and the *previous* event hash h_{prev}. The first event's predecessor is a fixed 64-zero genesis hash. The result is a linked chain whose unforgeability follows from the hash commitment: altering, deleting, or reordering any event changes its h_{event} and thus breaks every subsequent link and the signed head [Crosby and Wallach, 2009].

Canonicalization uses RFC 8785 JCS (UTF-16 property-name ordering, I-JSON safe-integer range) so that hashing is independent of key order or whitespace [Rundgren, 2020]. The device attests each signed high-water checkpoint with Ed25519 [Bernstein et al., 2012]; the fingerprint published in the manifest is SHA256 of the public key, and sensitive content is sealed with XChaCha20-Poly1305 [Bernstein, 2008].

4.3 Storage with crash recovery

Events are persisted as immutable, create-only encrypted `.segment` files named by zero-padded sequence, alongside an encrypted `objects/` store for captured artifacts and a `checkpoints/` directory of signed high-water markers (`signed-high-water/v1`). Each append writes the segment (with `fsync`), writes a *signed pending-append* record, then indexes and checkpoints, then removes the pending record. On open, the store reconciles any surviving pending append and rejects unsigned extra segments, missing or reordered segments, and truncation of the signed tail [Crosby and Wallach, 2009]. Legacy stores require an explicit, honestly bounded migration that seals validated local bytes without claiming to prove earlier existence.

4.4 Two-layer export and honest disclosure

Export produces a single Evidence Package v1 (a ZIP) with a signed manifest. The *public layer* contains `evidence.json` (events with payloads replaced by their hashes), a bilingual `report.html` (zh-CN/en), an English-summary `report.pdf`, and a `timestamp-target.json` for optional OpenTimestamps. The *sensitive layer* holds the full events, research items, artifacts, anchors, and AI disclosures, encrypted under a passphrase derived with Argon2id (Argon2id) [Biryukov et al., 2015]. The manifest carries an explicit `evidenceClaim` and a `limitations` list, and embeds a `capabilityReport` recording, per platform capability, one of Available, PermissionRequired, Degraded, or Unavailable.

This last point is the design's distinctive honesty mechanism. The macOS adapter is implemented; on Windows and Linux the same adapter reports capture as Unavailable rather than fabricating capability [Academic Integrity Recorder Project, 2026c]. Gaps are first-class evidence: explicit `Gap`/`Redaction` events record *why* coverage lapsed (permission denied, platform limitation, adapter failure, user pause/exclusion, content deleted/redacted, data unavailable). Manuscript anchors bind a research claim to a quote in a PDF/DOCX/TeX/Markdown source by SHA256 of the quote and surrounding context, and can be revalidated as Valid, Relocatable, or Stale [Academic Integrity Recorder Project, 2026b]. An *active-time* algorithm reports unique observed foreground activity, capped by a timeout and reset on pause, lock, sleep, or backgrounding—a conservative lower bound, never an over-claim.

4.5 Offline verification

A skeptic runs the verifier on the package alone. It checks the manifest Ed25519 signature, the final checkpoint signature, continuity of the public hash chain, every file's size and SHA256 digest against the manifest, and—given the sensitive-layer passphrase—that payload hashes reproduce, the event chain is complete, active time is consistent, and declared relationships hold. It rejects duplicate or undeclared ZIP entries, digest mismatches, and wrong passphrases. Success proves *process integrity and a device signature only*; the report states this in its output.

5 Scope, Limitations, Governance, and Adoption

5.1 Scope and what the record can show

Unforgeable process evidence is most valuable where the contested question is *how*, not *whether*: disclosure of AI assistance, provenance of a passage, the trail of revision, the existence of a planned experiment that was later dropped, or a contemporaneous lab record. It strengthens theses, grant reports, and post-publication audit alike.

5.2 Honest limitations and adversarial considerations

The system makes no claim to completeness or proof of integrity, and the verifier says so. Specifically:

- **Coverage upper bound.** Only observed activity is recorded. Offline work, use of an uninstrumented machine, or delegation to a colleague creates gaps the record cannot fill, and the design records these as gaps rather than hiding them [Academic Integrity Recorder Project, 2026d].
- **Platform dependence.** Capture fidelity is bounded by what the OS permits; where an adapter is UNAVAILABLE, the record says so [Academic Integrity Recorder Project, 2026a].
- **Conditional on assumptions.** Unforgeability is computational and conditional: it holds only while signing keys remain secret and any external timestamp is honest. A compromised key or a controlled clock can retrospectively undermine the record.
- **Not identity.** A device signature attests a key, not a legal person. It cannot prevent one person from using another's device or account.
- **Adversarial bypass.** A determined adversary can record on a separate machine, disable capture, or fabricate inputs before recording; the system raises the cost and leaves traces (gaps, inconsistencies) but is not a tamper-proof oracle.
- **Not a substitute for peer review.** The record is evidence to weigh, not a verdict on merit or originality.

These limits are features when stated plainly: an unforgeable record is trustworthy *because* it is honest about where it stops.

5.3 Governance and adoption path

We propose unforgeable process evidence as a lightweight, open community standard, in the spirit of how code signing and transparency logs became default expectations [Crosby and Wallach, 2009]. Concrete steps:

1. **Open specification and interop.** The EVIDENCE PACKAGE V1 format and verifier are open; journals or archives can run the verifier on deposited packages without vendor lock-in [Academic Integrity Recorder Project, 2026b,c].
2. **Complement, don't replace.** Tie the record to existing norms: the data-availability statement, the preregistration, and the conflicts/disclosure form. The AI-disclosure entity maps directly onto emerging AI-use policies [Gao et al., 2023, Committee on Publication Ethics (COPE), 2023].

3. **Tiered adoption.** Start with self-archiving and reviewer-on-request, then optional journal encouragement, then (where appropriate) a deposition requirement for certain claim types. Voluntary, low-friction adoption avoids the surveillance objection.

4. **Incentives for honesty.** Because gaps are recorded, a complete, gap-light record becomes a positive signal; a record full of disclosed gaps is at least honest, which is more than the current default of silence.

6 Conclusion

Reproducibility verifies *what* was claimed; unforgeable process evidence verifies *how* the claim was arrived at. We have argued that the second is a new paradigm of scientific integrity—not a replacement for reproducibility but a partner to it, closing the gap that reproducibility, by construction, cannot see. The property is attainable today with local, privacy-preserving, cryptographic recording and offline adversarial verification, as Academic Integrity Recorder and the Evidence Package v1 demonstrate, and it is honest about where it stops: it attests process integrity, never truth, originality, or authorship. Adopting unforgeable process evidence will not end the reproducibility crisis, but it gives the scientific community a new instrument for the part of integrity that reproducibility was never designed to observe.

Acknowledgments

The author thanks the open-source and open-science communities whose tools and norms this work builds upon. The reference implementation is released under the AGPL-3.0 license.

References

Academic Integrity Recorder Project. Platform capability matrix and honest-disclosure model. Repository documentation: `docs/PLATFORM_CAPABILITIES.md`, 2026a. URL https://github.com/chongliuresearch/academic-integrity-recorder.

Academic Integrity Recorder Project. Evidence package v1 specification. Repository documentation: `spec/evidence-package-v1.md`, 2026b. URL https://github.com/chongliuresearch/academic-integrity-recorder.

Academic Integrity Recorder Project. Capture adapter specification and platform capability model. Repository documentation: `spec/` and `docs/PLATFORM_CAPABILITIES.md`, 2026c. URL https://github.com/chongliuresearch/academic-integrity-recorder.

Academic Integrity Recorder Project. Threat model and non-claims. Repository documentation: `docs/THREAT_MODEL.md`, 2026d. URL https://github.com/chongliuresearch/academic-integrity-recorder.

Monya Baker. 1,500 scientists lift the lid on reproducibility. *Nature*, 533:452–454, 2016. doi: 10.1038/533452a.

C. Glenn Begley and Lee M. Ellis. Drug development: Raise standards for preclinical cancer research. *Nature Reviews Drug Discovery*, 11:733–735, 2012. doi: 10.1038/nrd3808.

Daniel J. Bernstein. ChaCha, a variant of Salsa20. eSTREAM portfolio submission, 2008. URL https://cr.yp.to/chacha/chacha-20080128.pdf.

Daniel J. Bernstein, Niels Duif, Tanja Lange, Peter Schwabe, and Bo-Yin Yang. High-speed high-security signatures. In *Proceedings of CHES 2012*, pages 124–142, 2012. doi: 10.1007/978-3-642-23951-9_9.

Alex Biryukov, Daniel Dinu, and Dmitry Khovratovich. Argon2: the memory-hard function for password hashing and other applications. PHC winner specification, 2015. URL https://www.password-hashing.net/argon2-specs.pdf.

Committee on Publication Ethics (COPE). ChatGPT and ai tools in manuscript preparation. COPE position statement, 2023. URL https://publicationethics.org/ai-author-guidelines.

Scott A. Crosby and Dan S. Wallach. Efficient data structures for tamper-evident logging. In *Proceedings of the 18th USENIX Security Symposium*, pages 317–334, 2009.

Timothy M. Errington, Maya Mathur, Courtney K. Soderberg, et al. Investigating the replicability of preclinical cancer biology. *eLife*, 10:e67401, 2021. doi: 10.7554/eLife.67401.

Daniele Fanelli. How many scientists fabricate and falsify research? a systematic review and meta-analysis of survey data. *PLoS ONE*, 4(5):e5738, 2009. doi: 10.1371/journal.pone.0005738.

Daniele Fanelli. Negative results are disappearing from most disciplines and countries. *Scientometrics*, 90(3):891–904, 2011. doi: 10.1007/s11192-011-0494-7.

Catherine A. Gao, Fred E. Howard, Nikolaus S. Markov, et al. Comparing scientific abstracts generated by ChatGPT to original abstracts using blinded human reviewers. *JAMA Network Open*, 6 (2):e230178, 2023. doi: 10.1001/jamanetworkopen.2023.0178.

Andrew Gelman and Eric Loken. The statistical crisis in science. *American Scientist*, 102(6):460–465, 2014.

John P. A. Ioannidis. Why most published research findings are false. *PLoS Medicine*, 2(8):e124, 2005. doi: 10.1371/journal.pmed.0020124.

Leslie K. John, George Loewenstein, and Drazen Prelec. Measuring the prevalence of questionable research practices with incentives for truth telling. *Psychological Science*, 23(5):524–532, 2012. doi: 10.1177/0956797611430953.

Francis L. Macrina. *Scientific Integrity: Text and Cases in Responsible Conduct of Research*. ASM Press, 5 edition, 2017.

Luc Moreau and Paul Groth. *Provenance: An Introduction to PROV*. Synthesis Lectures on the Semantic Web. Morgan & Claypool, 2015. doi: 10.2200/S00660ED1V01Y201410WBE013.

Luc Moreau and Paolo Missier. Prov-dm: The prov data model. W3C Recommendation, 30 April 2013, 2013. URL https://www.w3.org/TR/prov-dm/.

Brian A. Nosek, George Alter, George C. Banks, et al. Promoting an open research culture. *Science*, 348(6242):aab2374, 2015. doi: 10.1126/science.aab2374.

Brian A. Nosek, Charles R. Ebersole, Amanda C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018. doi: 10.1073/pnas.1708274114.

Open Science Collaboration. Estimating the reproducibility of psychological science. *Science*, 349 (6251):aac4716, 2015. doi: 10.1126/science.aac4716.

OpenTimestamps. Scalable, trust-minimized, distributed timestamping. Project documentation, 2018. URL https://opentimestamps.org/.

Florian Prinz, Thomas Schlange, and Khusru Asadullah. Believe it or not: how much can we rely on published data? *Nature Reviews Drug Discovery*, 10:712, 2011. doi: 10.1038/nrd3439-712b.

Anders Rundgren. JSON canonicalization scheme (JCS). IETF RFC 8785, 2020. URL https://www.rfc-editor.org/rfc/rfc8785.

Victoria Stodden. The scientific method: Reproducible research. *Computing in Science & Engineering*, 11(1):9–11, 2009. doi: 10.1109/MCSE.2009.14.

Paul Voigt and Axel Von dem Bussche. *The EU General Data Protection Regulation (GDPR): A Practical Guide*. Springer, 2017.

World Wide Web Consortium. Prov model primer. W3C Working Group Note, 30 April 2013, 2013. URL https://www.w3.org/TR/prov-primer/.